

Tutoring 01

SQL & Introduction to the WebApp Project

Francesco L. De Faveri

Web Applications Tutoring
Academic Year: 2025-2026



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~\$ whoami



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2018 - 2021



UNIVERSITÀ
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DI TRIESTE

**Bachelor Degree in
Mathematics
@ University of Trieste**

Thesis:

"A possibilistic Approach to
Fuzzy Arithmetic."

2021 - 2023



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DI PADOVA

**Master Degree in
Cybersecurity
@ University of Padova**

Thesis:

"A Feasibility Study and Privacy
Analysis of a Data Management
Infrastructure in the Oncology
Research Domain."



2023 - 20...



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MAX PLANCK INSTITUTE
FOR SECURITY AND PRIVACY



**PhD in Information Engineering
@ University of Padova
Visiting PhD
@ Max Planck Institute for Security and Privacy**

Research Areas:

Privacy Preserving Information Retrieval
Anonymization & Differential Privacy for structured and unstructured
data in Oncological Domains
Evaluation in Information Retrieval, Large Language Models

Outline

- General Tutorship Information
- Tutorship Project Description
- Data Layer
- Homework 1 - Overview



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Tutorship General Information



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Tutorship?

Main Goals:

- **Apply the concepts explained** by the professor during lectures
- Show you the development of a **web application step-by-step** (what you should do with your team)
- Discuss your **doubts/problems/questions** for the project

Feel free to ask questions related to your homeworks.



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Lectures

- Lectures will be on **Monday, 12:30-13:30, Room Ie**
- March
 - 09, **16***, 23, 30 (* next time will be in Room Me)
- April
 - 06, 20, 27
- May
 - 04, 11, 18



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Tutorship Roadmap

~5 lectures on Backend

- **09.03** -> Data Layer
- 16.03 -> Git & Maven etc.
- 23.03 -> Servlets
- 30.03 -> JSPs
- 06.04 -> MIME

~5 lectures on Frontend

- 20.04 -> REST / AJAX

HW1 Deadline on 24.04

- 27.04 -> HTML
- 04.05 -> CSS
- 11.05 -> JS 1
- 18.05 -> JS 2

HW2 Deadline on 05.06



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Useful links

Code examples:

<https://bitbucket.org/frncl/webapp-unipd/src/master/>

Past Year Tutorship:

A.Y. 2021/2022: <https://bitbucket.org/frncl/tutor-2022/src/master/>

A.Y. 2022/2023: <https://bitbucket.org/frncl/tutor-2023/src/master/>

A.Y. 2023/2024: <https://bitbucket.org/frncl/tutor-wa-2024/src/master/>

A.Y. 2024/2025: <https://bitbucket.org/frncl/tutor-wa-2025/src/master/>

(A. Y. 2024/2025 also for this year)



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Deadlines

Close deadlines:

- **CRANE: HW Groups Deadline:** Friday 13 March 2026!
- **HW1 Deadline:** Thursday 24 April 2026

Suggestion: do not limit yourself in finding colleagues to form a group, **discuss also about the web app** that you will develop (and **talk with the professor** about your group idea)



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Homework 1

Deadline: Thursday 24 April 2026

LaTeX Template: <https://bitbucket.org/frncl/wa-homework-template/src/master/>

We suggest to use LaTeX there exist many plug-ins for your preferred IDEs:

IntelliJ: <https://plugins.jetbrains.com/plugin/9473-texify-idea>

VS-code:

<https://marketplace.visualstudio.com/items?itemName=James-Yu.latex-workshop>

You can also install a LaTeX distribution locally: <https://www.latex-project.org/get/>



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Overleaf? Probably not a good idea...

Overleaf



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What your Bitbucket Project Repo need

Remember, you need to **equally contribute** to every part of the web application including:

- Back-end code (server-side)
- Front-end code (client-side)
- Report (HW1)
- Slides & presentation

Your contribution are also **checked through the commit history** of your bitbucket repository, thus:

- **Commit regularly the code** that you produce
- Also for the **LaTeX project** commit regularly



Tutorship Project



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Tutorship Project - SIGIR WebApp ([ACM SIGIR 25](#))



SIGIR is A* international conference for Information Retrieval.

Those papers are categorized based on the track to which they are submitted:

- Full
- Short
- Resource & Reproducibility
- Perspectives
- Workshops
- Tutorials

The submitted papers are then reviewed and accepted/rejected.
The accepted papers must be presented during the SIGIR conference.

Sunday, July 13, 2025 TUTORIALS DAY FROM 8:00 - REGISTRATIONS 9:00 - 10:30 Tutorials • R²LLMs: Retrieval and Ranking with LLMs • Lona Context vs. RAG: Strategies	Monday, July 14, 2025 MAIN CONFERENCE DAY 1 FROM 7:30 - REGISTRATIONS 8:30 - 9:00 Conference Opening 9:00 - 10:00	Tuesday, July 15, 2025 MAIN CONFERENCE DAY 2 FROM 8:30 - REGISTRATIONS 9:00 - 10:00 Keynote 2: Ophir Frieder: Digital Health 10:00 - 10:30
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Tutorship Project - SIGIR WebApp

The main functions of the Web App should help us to:

- **Collect the presentations** (slides) related to the papers.
- **Present the data associated to the papers and of the related authors** in a suitable and clear way.
- Allows to **keep track of the conference “program”**, that is the time, location of the various presentations and events.



Tutorship Project - SIGIR WebApp

Structure:

- Data Layer (**PostgreSQL**)
- Business Logic Layer (**Java Servlets**)
- Presentation Layer (**HTML5, CSS3, JS**)



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Data Layer



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Data Layer

The database represents the foundations of your web application:

- It is important to **carefully design the database** to have full control over the data, independently of the other layers.
- Note that if the database is NOT well designed, a change in the database may lead to **major changes to the other layers** (or the other way around) and **security problems**.
- It is important to **know how to interact with the database**. This allows to better understand the behaviour of your web app.



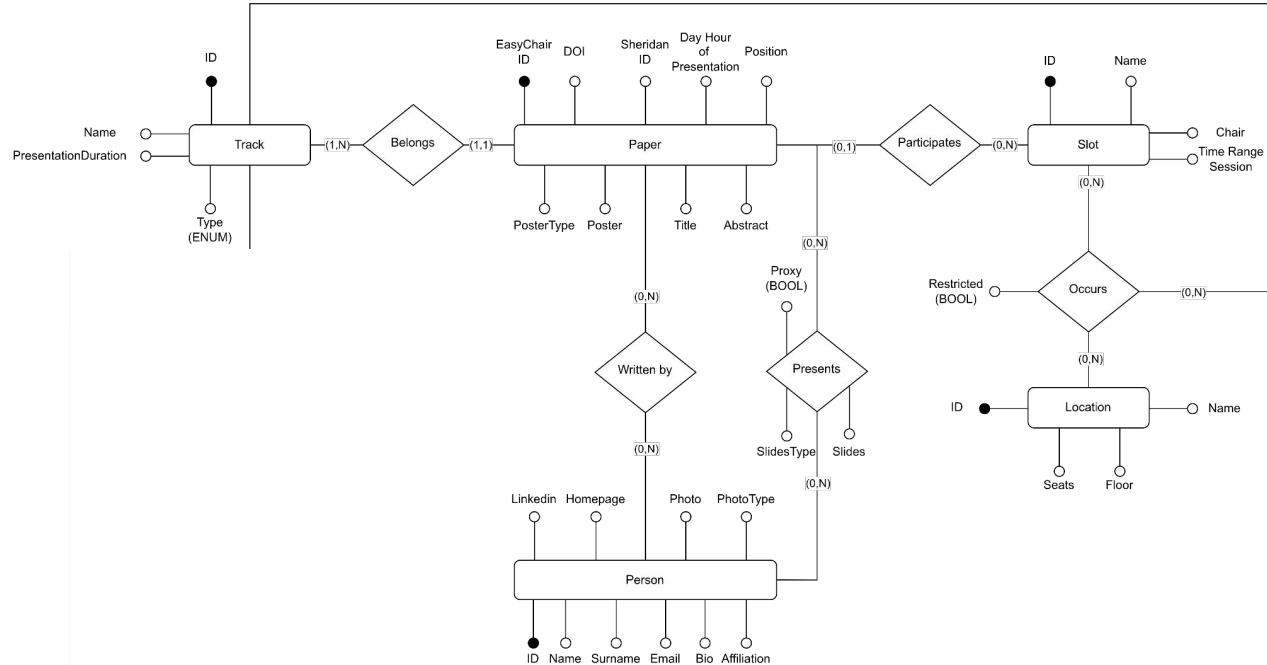
Data Layer

Therefore, the database that you design for your homeworks should be:

- **Reasonably complex** (but not too much): The DB should allow you to solve all the real problems for the situation you are designing the web app.
- **Consistent:** Inconsistencies may lead to challenges in the development.

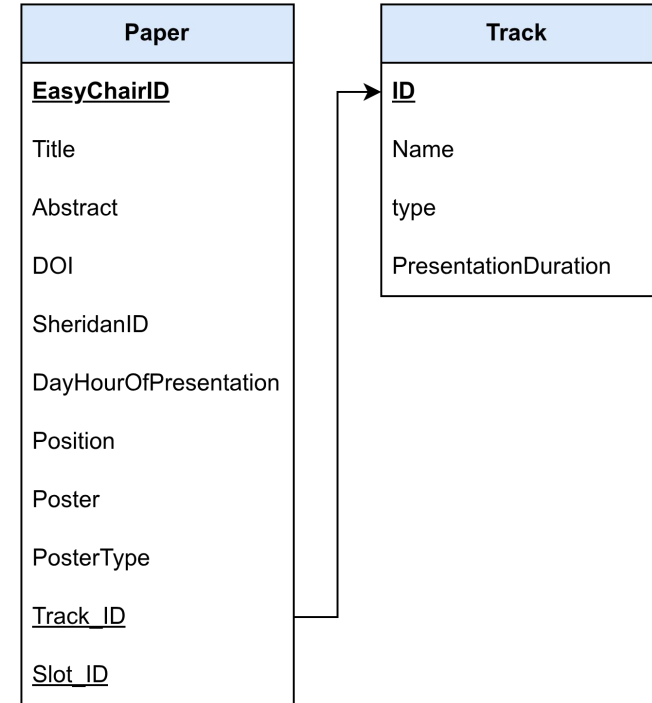
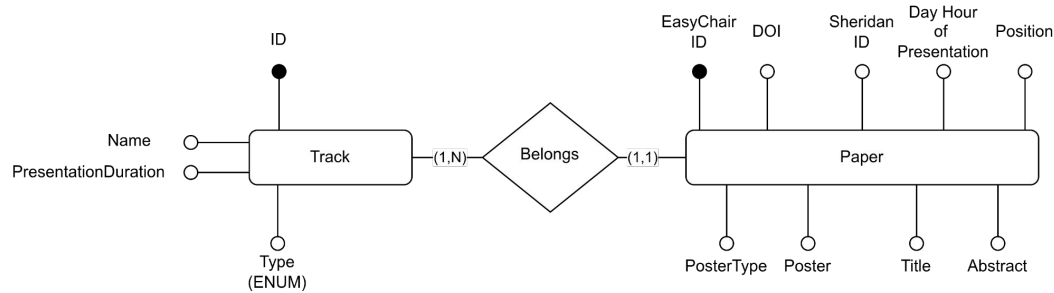


SIGIR WebApp - ER schema



SIGIR WebApp - ER schema

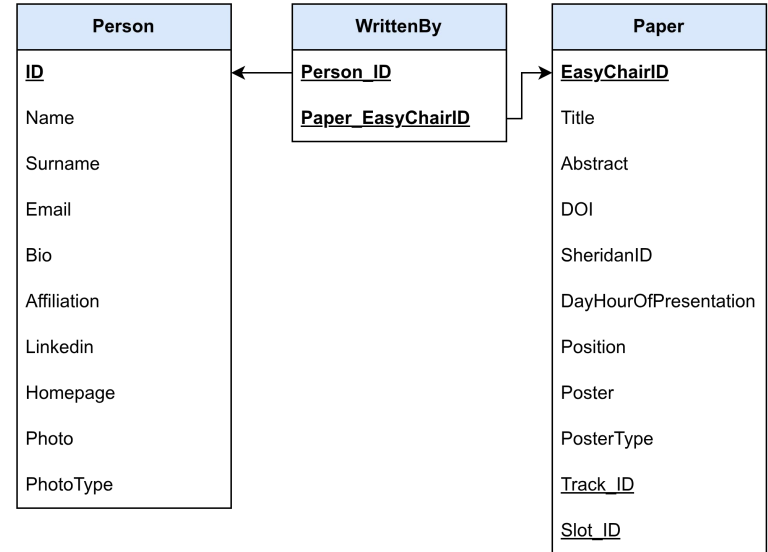
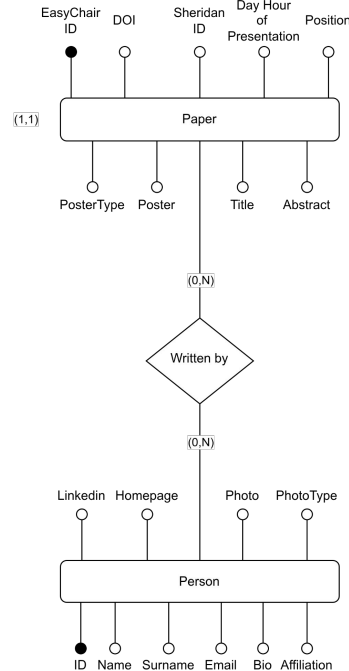
A Paper belongs to a Track



SIGIR WebApp - ER schema

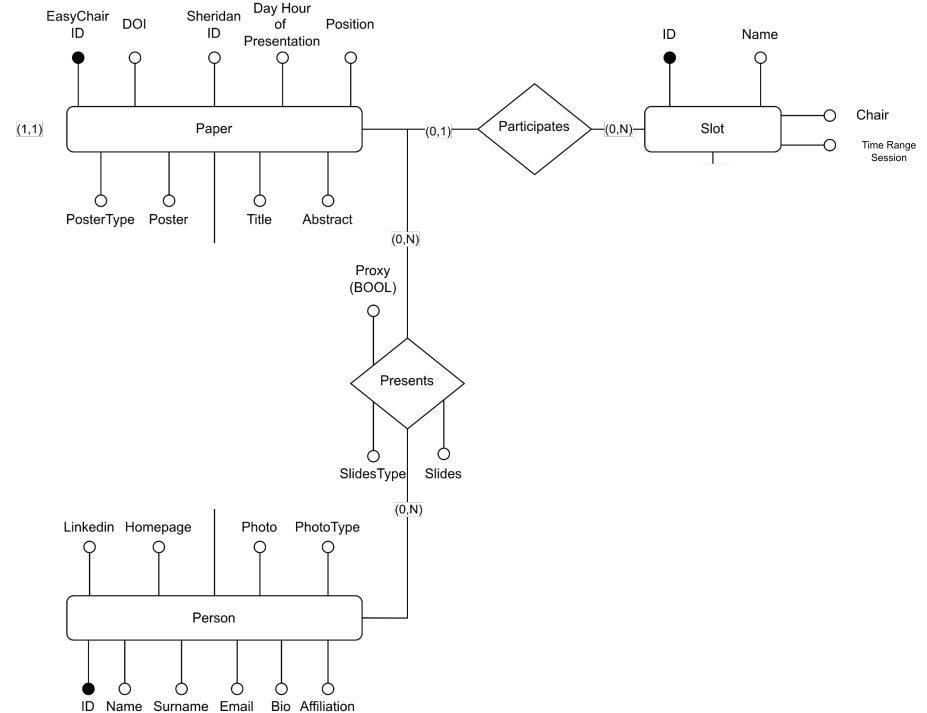
A Paper is written by a Person (Author)

Remark: The cardinality of the Authors is (0,N) because there are some special cases.



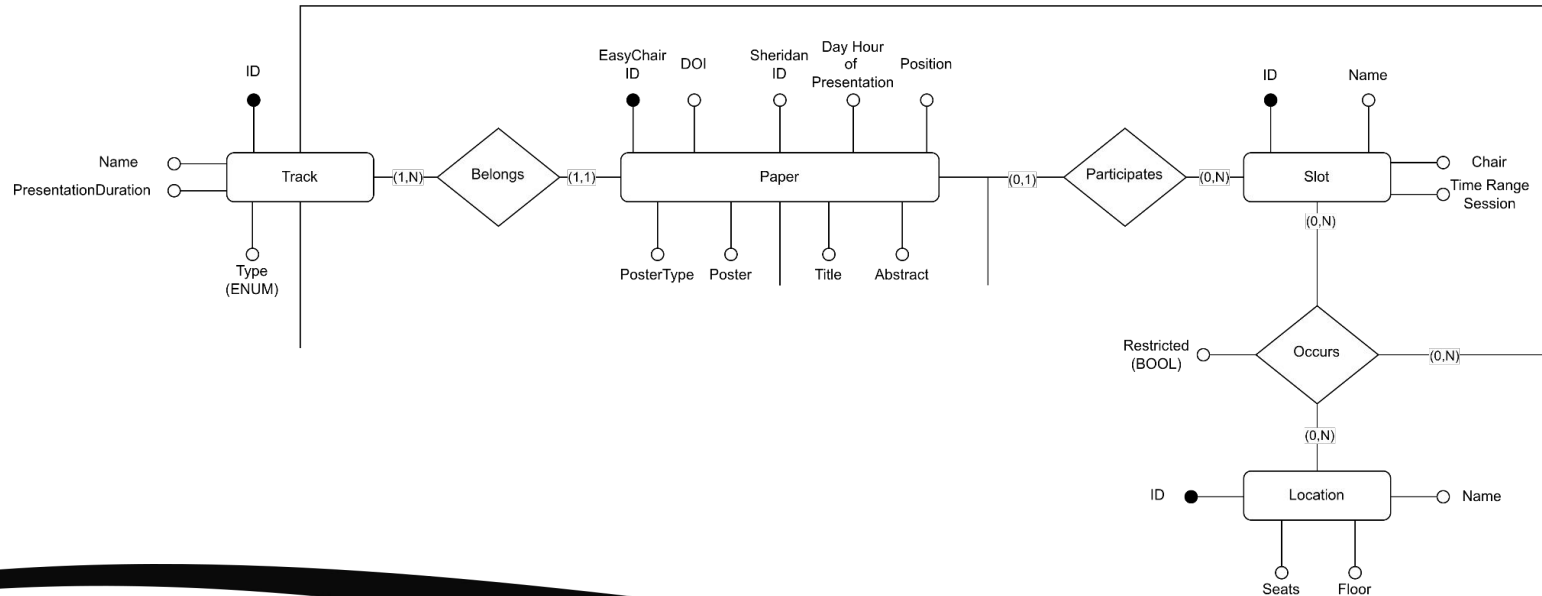
SIGIR WebApp - ER schema

A Paper is presented by a Person in a certain time slot.

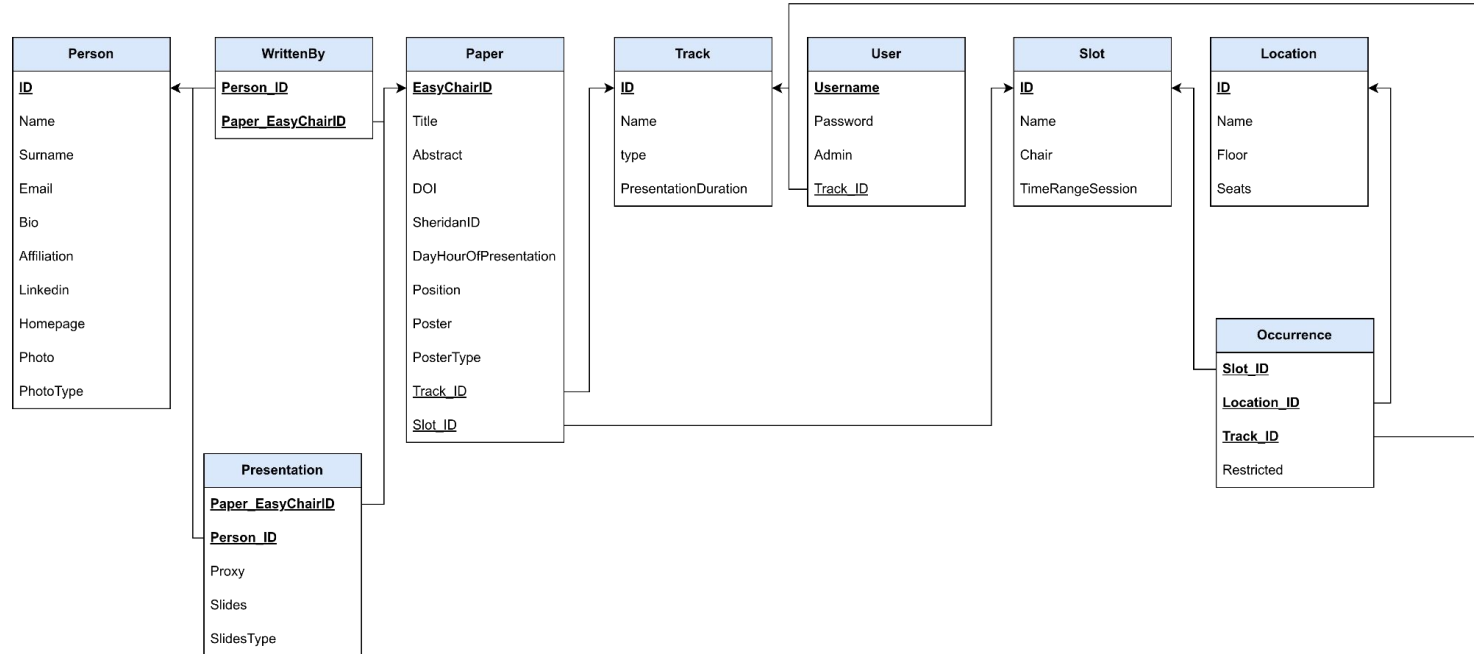


SIGIR WebApp - ER schema

A Slot is associated to both a Track and a Location. The “Loop” Track->Paper->Slot->Track is needed because there may be **slots in which no paper is presented** (e.g., **Social Events**).



SIGIR WebApp - Logical Model



SIGIR WebApp - Physical Model

```
40  ⚠ Drop the Webappsigir schema, if exists, and any object within it
41  DROP SCHEMA IF EXISTS Webappsigir CASCADE;
42  DROP TYPE IF EXISTS TRACK_TYPE CASCADE;
43
44
45  CREATE SCHEMA Webappsigir;
46  COMMENT ON SCHEMA Webappsigir IS 'Schema for SIGIR webapp';
47
48
49  -- #####
50  -- ## Creation of the tables
51  -- #####
52
53
54  --
55  -- This table represents a person (author or presenter of a paper)
56  --
57  -- Version 1.00
58  CREATE TABLE Webappsigir.Person (
59      id SERIAL,
60      name VARCHAR NOT NULL,
61      surname VARCHAR NOT NULL,
62      bio VARCHAR,
63      email VARCHAR NOT NULL,
64      affiliation VARCHAR NOT NULL,
65      linkedin VARCHAR,
66      homepage VARCHAR,
67      photo BYTEA,
68      photo_type TEXT,
69      PRIMARY KEY (id)
70  );
```



PostgreSQL - Installation

PostgreSQL represents the **database server** which allows to create databases, perform queries and much more.

Download link:

<https://www.enterprisedb.com/downloads/postgres-postgresql-downloads>

During the installation you will be asked to select some components, make sure they are **ALL selected**.



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PostgreSQL - Installation

After the installation:

- windows→make sure that PostgreSQL bin folder has been added to the past system variable
- MacOS/Linux→after the installation, open a terminal and run:
export PATH=/Library/PostgreSQL/VERSION/bin:\$PATH

To verify your installation and be sure that postgres has been added to your paths run the command:

psql --version

It should return the installed version of PostgreSQL.

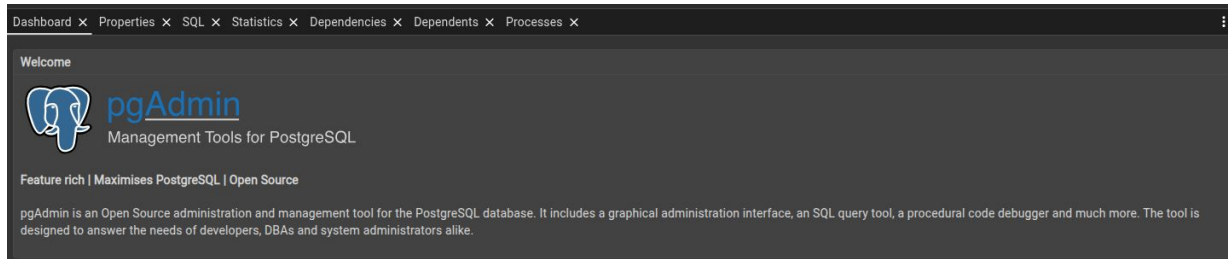


PostgreSQL - Managing DBs

To manage our databases we have 2 options:

psql: CLI, very useful when using (remote) servers

PgAdmin: GUI, allows interaction with your DB.



PostgreSQL - Managing DBs (psql)

Main psql commands

default: postgres

- **psql -U <username>**
- **\l \list**
- **\c <database_name>**
- **\dn**
- **\dt schema_name.***
- **\d <schema_name>.<table_name>**
- **\q**

Meaning

- Connect to PostgreSQL server
- List the databases
- Connect to database
- List the schemas
- List the tables of a schema
- Get table of a schema
- Quit



PostgreSQL - Managing DBs (psql)

- **sigir25** is the DB
- **webappsigir** is the schema
- ***** is used to display **all tables**

```
sigir25=> \dt webappsigir.*
```

List of tables

Schema	Name	Type	Owner
webappsigir	location	table	postgres
webappsigir	occurrence	table	postgres
webappsigir	paper	table	postgres
webappsigir	person	table	postgres
webappsigir	presentation	table	postgres
webappsigir	slot	table	postgres
webappsigir	track	table	postgres
webappsigir	user	table	postgres
webappsigir	writtenby	table	postgres

(9 rows)



PostgreSQL - Managing DBs (psql)

Dump the database:

- .sql files → **pg_dump -U postgres <db_name> <path_to_save .sql>**
- .tar files → **pg_dump -U postgres <db_name> <path_to_save .sql>**

Restore a database:

- .tar files → **pg_restore -d <db_name> -U postgres --verbose <path_to_tar>**
- .sql files → **psql -U postgres -d <db_name> -f <path_to_sql>**

Run a .sql script:

- **psql -U postgres -d <database_name> -f <path_to_sql>**



PostgreSQL - Managing DBs (psql)

To query the database you just need to connect to it. Then you can type the desired query.

Login to db sigir25,
server localhost

Run a simple query to see
the available locations.

```
psql -U postgres -d sigir25 -h localhost
Password for user postgres:
psql (18.3 (Ubuntu 18.3-1.pgdg24.04+1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off, ALPN: postgresql)
Type "help" for help.

sigir25=# SELECT * from webappsigir.location;
 id |      name      | floor | seats
----+-----+-----+-----
  1 | GIOTTO         |      0 | 1380
  2 | GIOTTO Platea  |      0 |  654
  3 | GIOTTO Galleria |      0 |  463
  4 | GIOTTO Balconata |      0 |  263
  5 | MANTEGNA       |      0 |  988
  6 | MANTEGNA Platea |      0 |  514
  7 | MANTEGNA 1     |      0 |  237
  8 | MANTEGNA 2     |      0 |  237
  9 | CARRARESI      |      0 |  120
 10 | ANTENORE A     |      0 |   50
 11 | ANTENORE B     |      0 |   50
 12 | DONATELLO A    |      0 |   50
 13 | DONATELLO B    |      0 |   50
 14 | PETRARCA A     |      0 |   50
 15 | PETRARCA B     |      0 |   50
 16 | PALLADIO A     |      0 |   50
 17 | PALLADIO B     |      0 |   50
(17 rows)

sigir25=#
```



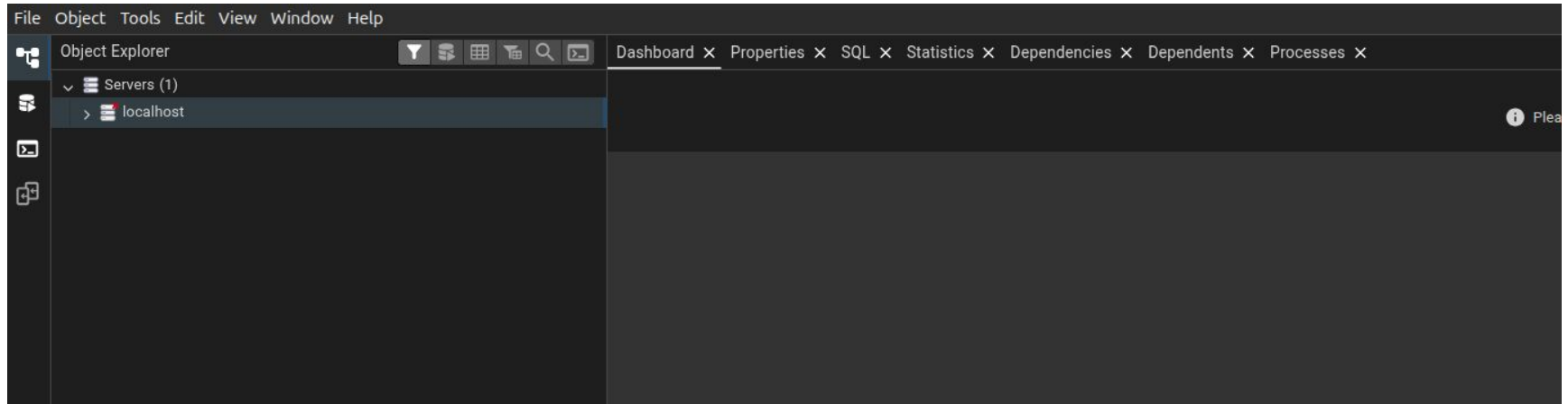
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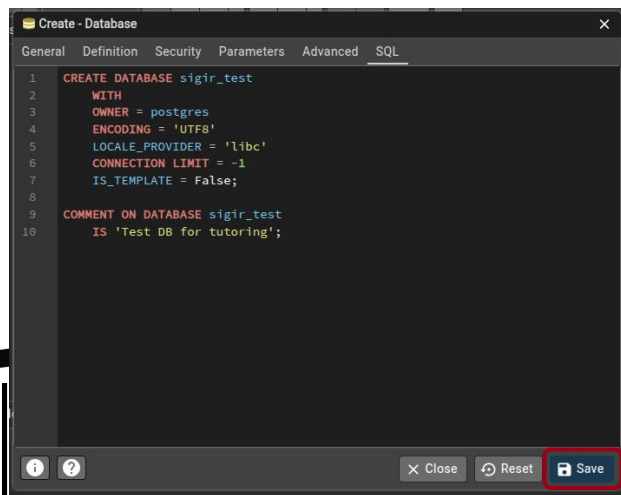
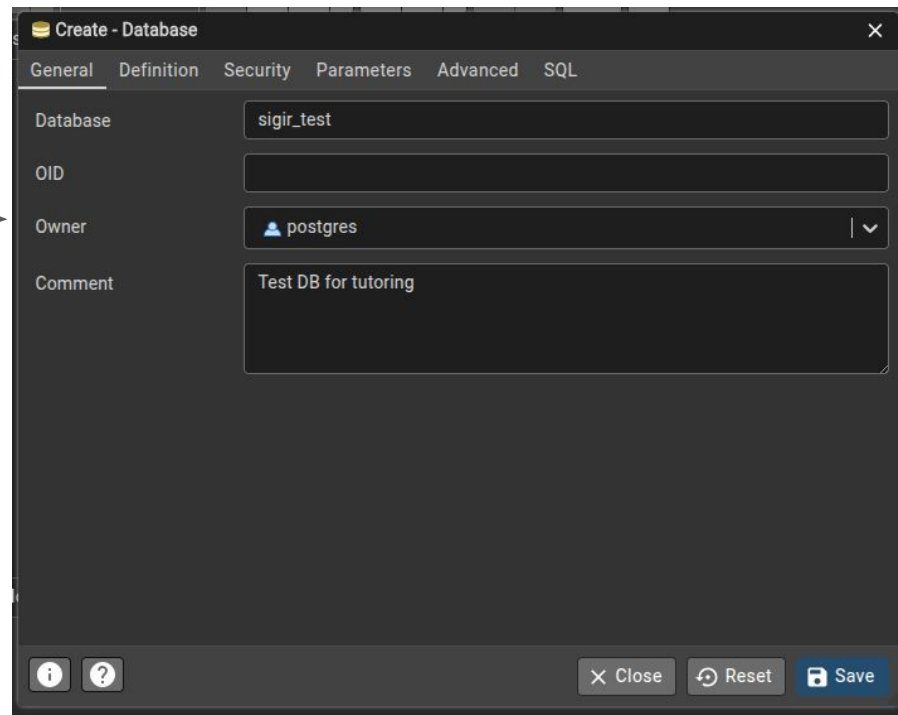
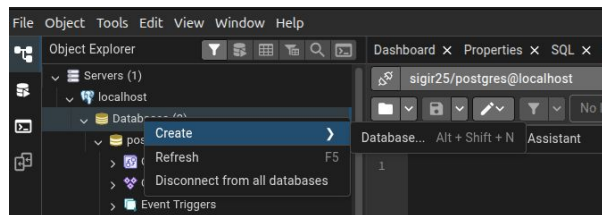
PostgreSQL - Managing DBs (PgAdmin)

Connect to server



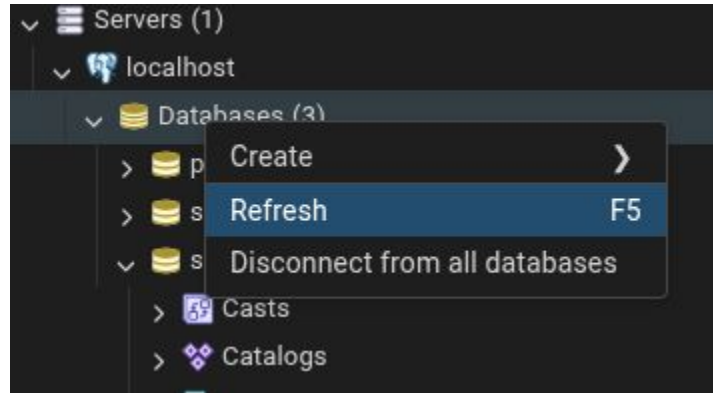
PostgreSQL - Managing DBs (PgAdmin)

Create Database



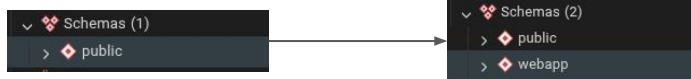
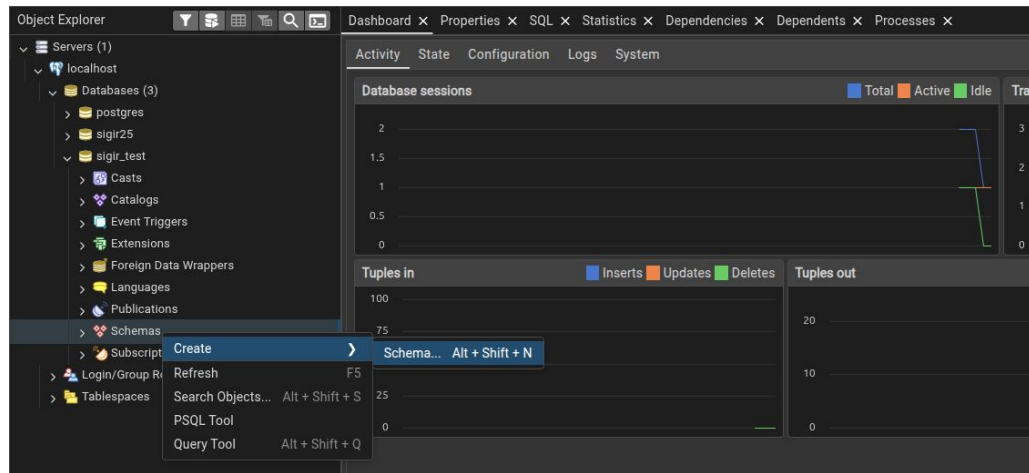
PostgreSQL - Managing DBs (PgAdmin)

It is possible that you are not able to see the DB immediately. If you do not see the database that you created: **refresh**

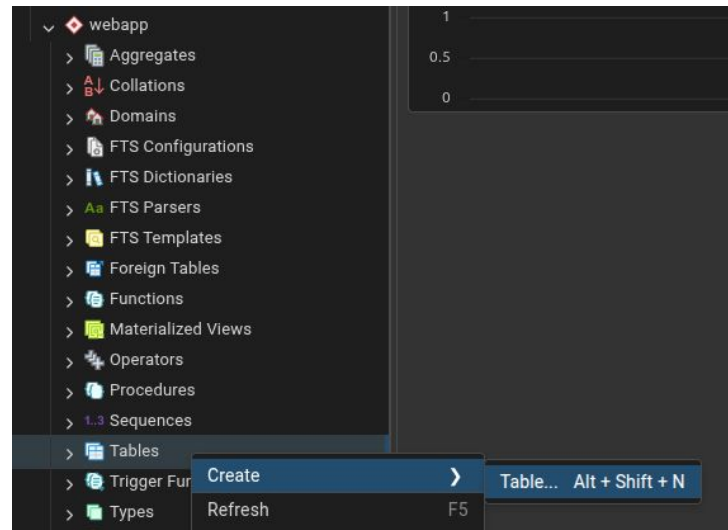


PostgreSQL - Managing DBs (PgAdmin)

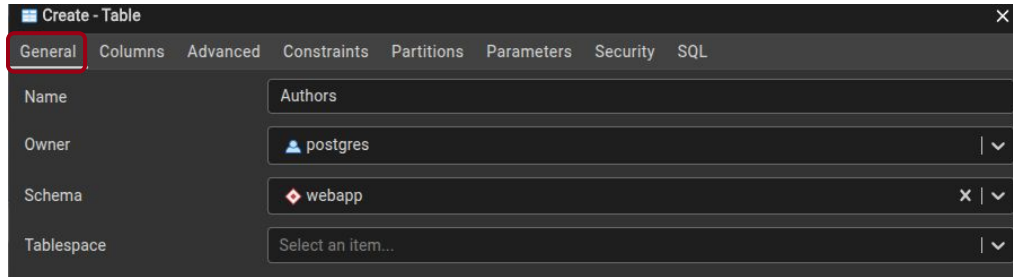
Create a schema



Create a table



PostgreSQL - Managing DBs (PgAdmin)

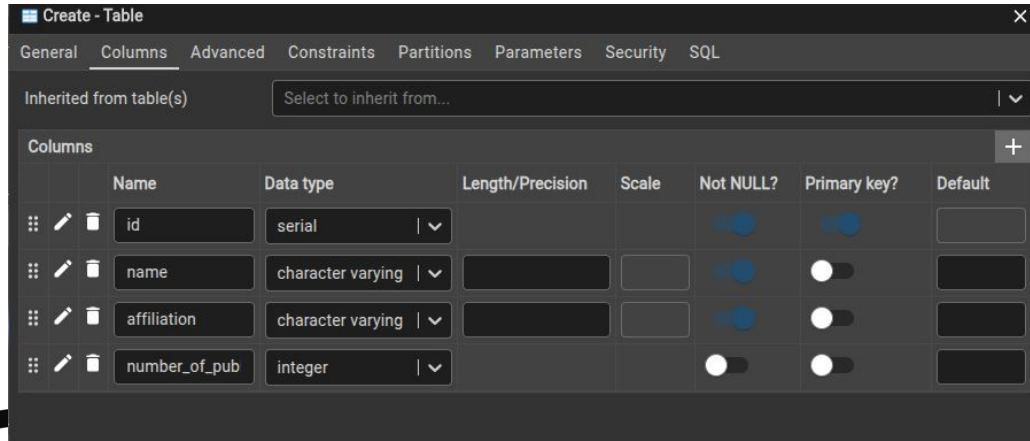


The screenshot shows the 'Create - Table' dialog box with the 'General' tab selected. The fields are as follows:

Field	Value
Name	Authors
Owner	postgres
Schema	webapp
Tablespace	Select an item...

In the interactive mode of creating a table you can set:

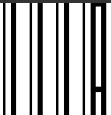
- name of the attribute (id, name, aff...)
- data type
- if NULL
- if Primary Key (PK)



The screenshot shows the 'Create - Table' dialog box with the 'Columns' tab selected. The 'Inherited from table(s)' field is set to 'Select to inherit from...'. The 'Columns' table is as follows:

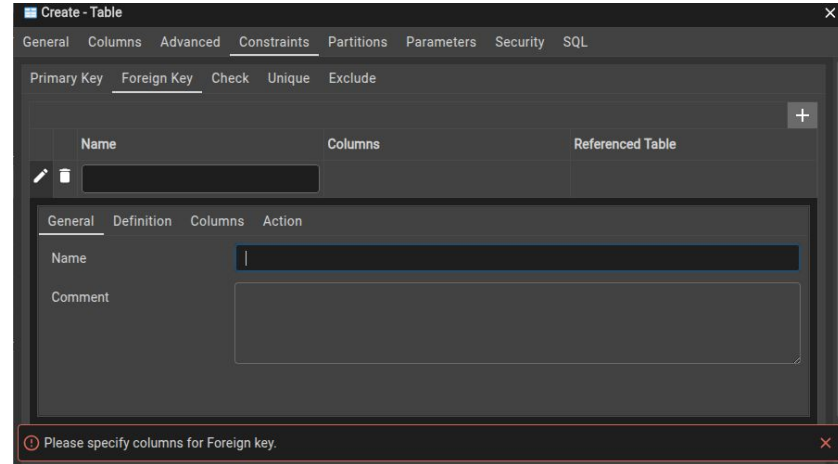
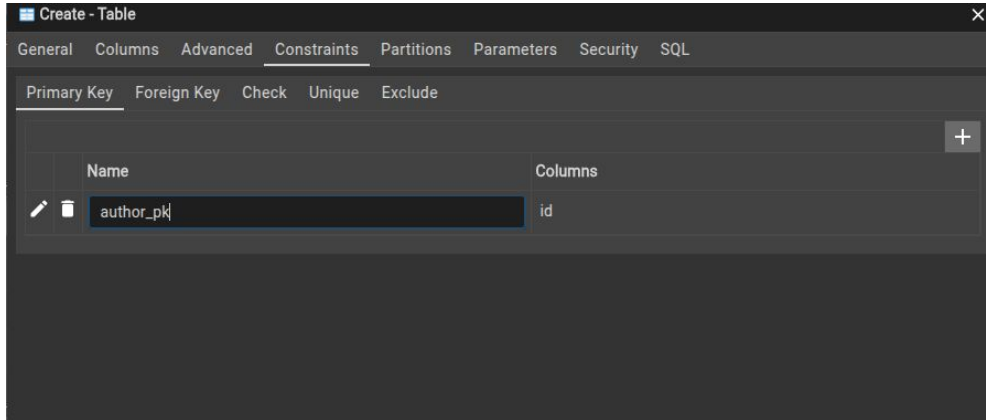
	Name	Data type	Length/Precision	Scale	Not NULL?	Primary key?	Default
☒	id	serial			☒	☒	
☒	name	character varying			☒	☐	
☒	affiliation	character varying			☒	☐	
☒	number_of_pub	integer			☐	☐	

To create table you can also use the query tool



PostgreSQL - Managing DBs (PgAdmin)

constraints: PK and FK

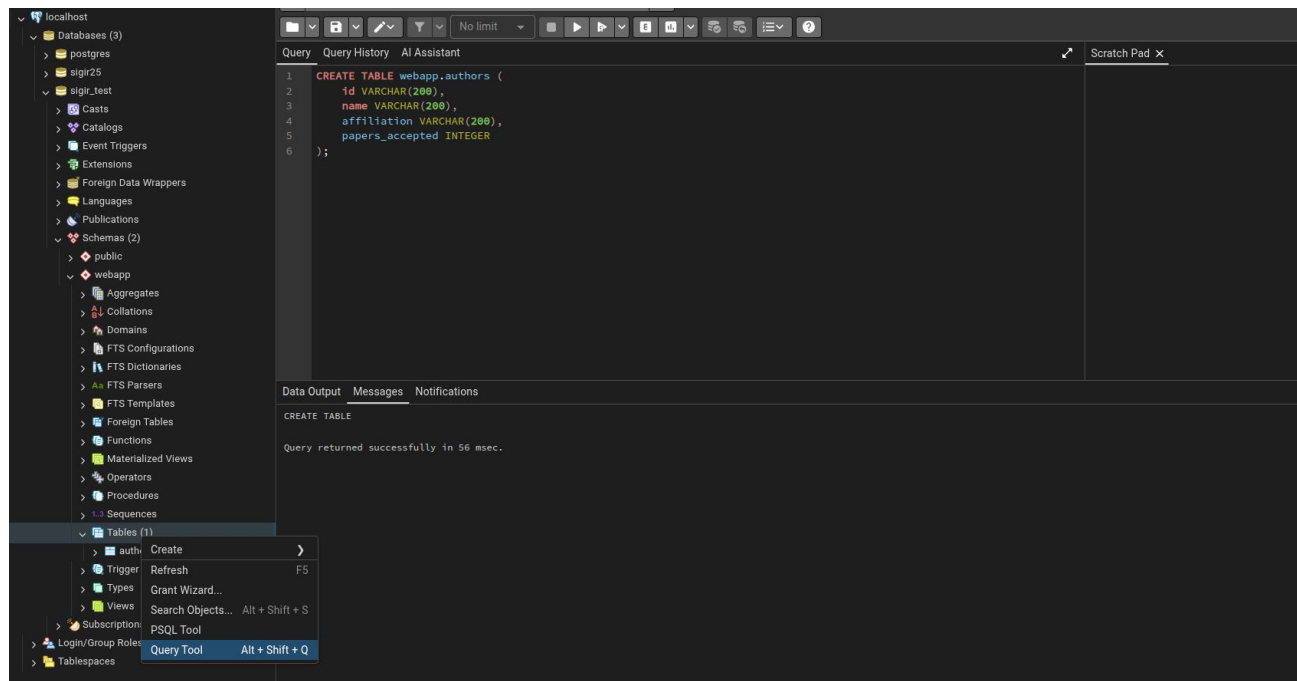


PostgreSQL - Managing DBs (PgAdmin)

querying

log messages
are displayed
below the shell

(remember to
refresh in case)



Homework 1 - Overview



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HW1 - Overview

Objectives:

Briefly describe the objectives of your web application. Why is it useful/important? Who are the users? Do not be too detailed in this section, be concise and effective at the same time.

Example:

SIGIR's webapp is a web application whose main goal is to ease the collection and management of the presentations (slides and posters) related to the papers accepted for publication in the context of the SIGIR conference. This project aims at providing authors with a platform suitable for uploading their data, including personal data and slides or posters; on the other hand it should allow to manage the presentations by keeping track of the presenters, the time and the location of conference events.



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HW1 - Overview

Main Functionalities

Describe in details what your web application is supposed to do, and what their users will be able to do. Try to be detailed in this section when you describe all the functionalities.

Example: The authors of a paper are allowed to upload their slides or posters that they use to present their paper. Moreover, they can upload their photo, bio and homepage/LinkedIn links which allow to provide more detailed information about an author. [...]



HW1 - Overview

Data Logic Layer

Describe the database of your web application. Describe the entities of your schema, providing, when needed, information about identifiers, attributes' data type.

Business Logic Layer

Report and describe the class diagram and the sequence diagram (you will see them in details in the next lessons)

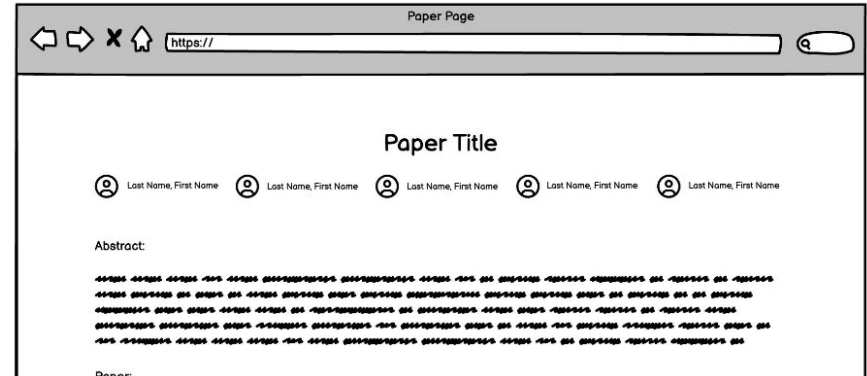


HW1 - Overview

Presentation Logic Layer

Describe the pages of your web applications (e.g., login page, authors' upload page, etc.). Describe each page in detail, providing information about the content of the web page, the information displayed, and what actions can be performed (e.g., insert new data about a paper). For each web page you plan to introduce, **provide also an interface mockup**, which is an image representing how you plan to implement the web page.

Suggestion: Balsamiq (<https://balsamiq.com/>)



HW1 - Overview

Group Members Contribution Section

Log everything you do as you do it. The professor looks at your commits and reads this section to evaluate the individual contribution of each one of you.

You are expected to **code**, **write** and **commit** everything in you personal repository along the project.

NOT TO DO:

- **Commit everything at the end** (One “Big” commit before the HW deadline)
- **Separation of labour in writing & coding teams**
(all of you needs to code and write for participating to the project)

